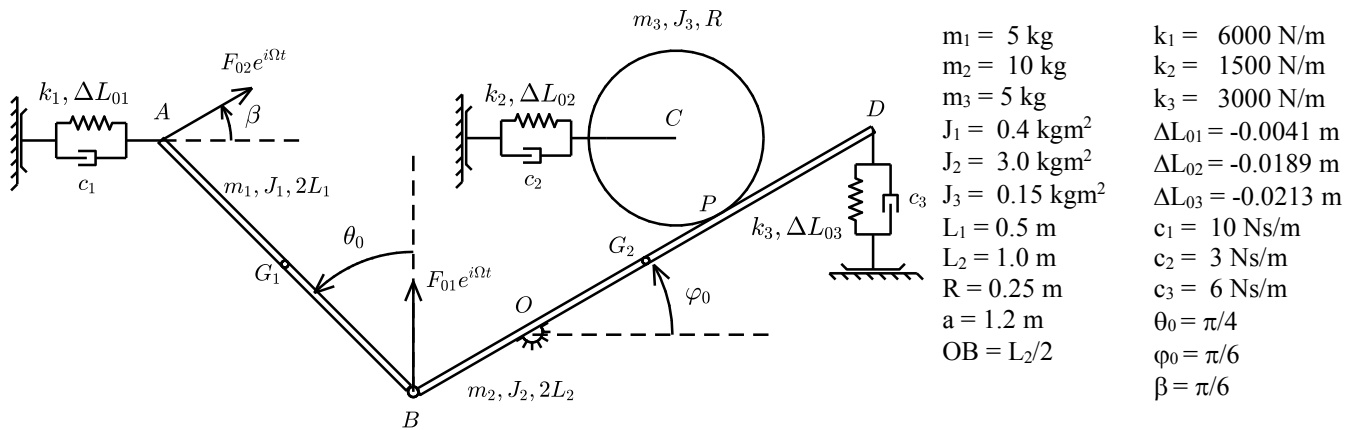


Dynamics of Mechanical Systems

January 11th, 2021



The mechanical system above lies in the vertical plane and is shown in its equilibrium position. The disk rolls without sliding on the rigid bar BD and the force applied to point A is inclined with respect to the horizontal direction by the constant angle β . In the equilibrium position, the distance between points O and P is $OP=a=1.2 \text{ m}$

The students are requested to:

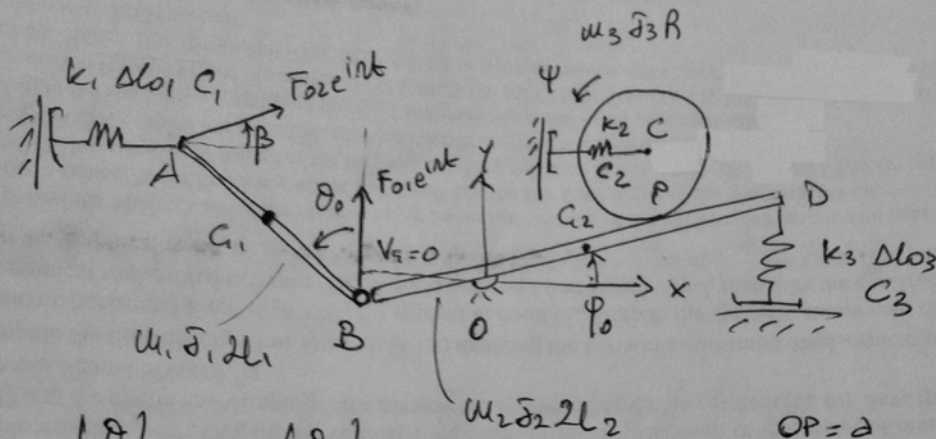
0. Write on top of each page their surname, name, ID number **and SNxxx code** (see below)
1. Write the linearized equations of motion around the given equilibrium configuration; in response to this point, upload on MS-Forms one single file in pdf format containing:
 - i. the drawing of the system marked with all kinematic variables used to solve the exam;
 - ii. The vector of independent coordinates chosen to write the system's equations of motion;
 - iii. The Jacobian matrices $[\Delta m]$, $[\Delta c]$, $[\Delta k]$, with evidence of the procedure used to derive these results;
 - iv. The gravitational & second order stiffness, with evidence of the procedure used to derive these results;
2. Compute the natural frequencies and modal shapes;
3. Compute the Frequency Response Function (in the range 0 – 10 Hz, step = 0.01 Hz) applying the harmonic force $F_{01}e^{i\Omega t}$. Output: the vertical component of the displacement of the center of the disk and the absolute rotation of the disk. Save the graphical results as SNxxx1.fig and SNxxx2.fig, with S the first letter of your surname, N the first letter of your name and xxx are the last three digits of your matriculation number (e.g. for Bruni Stefano 123456: BS4561.fig and BS4562.fig)
4. Compute the Frequency Response Function (in the range 0 – 10 Hz, step = 0.01 Hz) applying the harmonic force $F_{02}e^{i\Omega t}$. Output: the horizontal component of the displacement of the center of mass of the rigid bar AB and the elastic and damping force transmitted by the spring/damper system (k_3 , c_3). Save the graphical results as SNxxx3.fig and SNxxx4.fig
5. Compute the Frequency Response Function (in the range 0 – 10 Hz, step = 0.01 Hz) applying the harmonic force $F_{02}e^{i\Omega t}$. Output: the horizontal and vertical components of the constraint force in the hinge B. Save the graphical result as SNxxx5.fig SNxxx6.fig. In response to this point, also include in the pdf file uploaded on MS-Forms:
 - v. A drawing and the symbolic equations required to derive horizontal and vertical components of the constraint force in the hinge B .

The answer to points 2 – 5 shall be provided by delivering the MATLAB script and .fig files as a compressed folder uploaded on BeeP, see instructions below.

Note: Please collect the Matlab's script file relative to the numerical solution of the exercise and save it with name SNxxx.m, with S the first letter of the surname, N the first letter of the name and xxx the last three digits of the matriculation number (e.g. for Bruni Stefano 123456: BS456.m). It is mandatory that all of the files (fig e .m) are collected in one single compressed folder (in .zip or.rar format) named SNxxx.zip (or SNxxx.rar), which has to be uploaded on Beep. To upload the sipped folder:

1. Connect to the site <https://beep.metid.polimi.it> and enter the portal using your credentials;
3. Select section **Homework** of the course DYNAMICS OF MECHANICAL SYSTEMS (BRUNI STEFANO);
4. Select the folder **Test January 11th 2021 - standard**
5. Click on button Add and select Single document;
6. Browse the compressed file SNxxx.zip using the button Browse... (or Sfogli...
7. Fill the "Title" field with your code SNxxx (e.g. for Bruni Stefano 123456: BS456);
8. Confirm the procedure clicking the button Publish.

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$$\underline{x} = \begin{Bmatrix} \theta \\ \psi \\ \varphi \end{Bmatrix} \quad \underline{x}_0 = \begin{Bmatrix} \theta_0 \\ \psi_0 \\ 0 \end{Bmatrix}$$

ψ : relative rotation of the disk with respect to φ

$$x_B = -\frac{L_2}{2} \cos \varphi \quad v_{c2} = \frac{L_2}{2} \dot{\varphi}$$

$$y_B = -\frac{L_2}{2} \sin \varphi$$

$$x_{c1} = -\frac{L_2}{2} \cos \varphi - L_1 \sin \vartheta$$

$$y_{c1} = -\frac{L_2}{2} \sin \varphi + L_1 \cos \vartheta$$

$$x_p = (2 - R\psi) \cos \varphi$$

$$y_p = (2 - R\psi) \sin \varphi$$

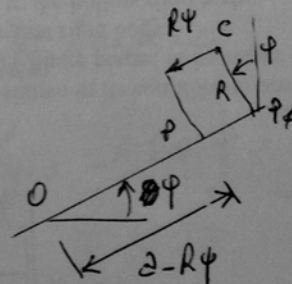
$$x_c = (2 - R\psi) \cos \varphi - R \sin \varphi$$

$$y_c = (2 - R\psi) \sin \varphi + R \cos \varphi$$

$$y_A = -\frac{L_2}{2} \sin \varphi + 2L_1 \cos \vartheta$$

$$x_A = -\frac{L_2}{2} \cos \varphi - 2L_1 \sin \vartheta$$

$$\Delta L_1 = x_A - x_{Ap} = -\frac{L_2}{2} \cos \varphi - 2L_1 \sin \vartheta - x_{Ap}$$



$$\Delta L_2 = x_c - x_{cp} = (2 - R\psi) \cos \varphi - R \sin \varphi - x_{cp}$$

$$y_D = \frac{3}{2} L_2 \sin \varphi$$

$$\Delta L_3 = y_D - y_{Dp} = \frac{3}{2} L_2 \sin \varphi - y_{Dp}$$

$$\psi_A = \psi + \varphi$$

$$= \frac{1}{2} m_1 \dot{x}_1^2 + \frac{1}{2} m_1 \dot{y}_1^2 + \frac{1}{2} J_1 \dot{\theta}^2 + \frac{1}{2} m_2 v_{12}^2 + \frac{1}{2} J_2 \dot{\varphi}^2 +$$

$$+ \frac{1}{2} m_3 \dot{x}_c^2 + \frac{1}{2} m_3 \dot{y}_c^2 + \frac{1}{2} J_3 \dot{\psi}^2$$

$$D = \frac{1}{2} \sum_{i=1}^3 C_i \Delta l_i^2; \quad V_{elc} = \frac{1}{2} \sum_{i=1}^3 k_i \Delta l_i^2; \quad V_g = m_1 g h_{g1} + m_2 g h_{g2} + m_3 g h_{g3}$$

$$\delta W = F_{01} e^{int} \delta y_B + F_{02} e^{int} \cos \beta \delta x_A + F_{02} e^{int} \sin \beta \delta y_A$$

$$[m_{PH}] = \text{diag}(m_1, m_1, J_1, m_2, J_2, m_3, m_3, J_3)$$

$$[C_{PH}] = \text{diag}(C_1, C_2, C_3); \quad [k_{PH}] = \text{diag}(k_1, k_2, k_3)$$

$$\underline{F} = \begin{Bmatrix} F_{01} & F_{02} \cos \beta & F_{02} \sin \beta \end{Bmatrix}^T e^{int}$$

$$\frac{\partial x_{g1}}{\partial \theta} = -L_1 \cos \theta \quad \frac{\partial x_{g1}}{\partial \varphi} = \frac{L_2}{2} \sin \varphi$$

$$\frac{\partial y_{g1}}{\partial \theta} = -L_1 \sin \theta \quad \frac{\partial y_{g1}}{\partial \varphi} = -\frac{L_2}{2} \cos \varphi$$

$$\frac{\partial x_c}{\partial \varphi} = -(2-R\psi) \sin \varphi - R \cos \varphi; \quad \frac{\partial x_c}{\partial \psi} = -R \cos \varphi$$

$$\frac{\partial y_c}{\partial \varphi} = (2-R\psi) \cos \varphi - R \sin \varphi; \quad \frac{\partial y_c}{\partial \psi} = -R \sin \varphi$$

$$\frac{\partial \Delta l_1}{\partial \theta} = -2L_1 \cos \theta \quad \frac{\partial \Delta l_1}{\partial \varphi} = \frac{L_2}{2} \sin \varphi$$

$$\frac{\partial \Delta l_2}{\partial \varphi} = -(2-R\psi) \sin \varphi - R \cos \varphi; \quad \frac{\partial \Delta l_2}{\partial \psi} = -R \cos \varphi$$

$$\frac{\partial \Delta l_3}{\partial \varphi} = \frac{3}{2} L_2 \cos \varphi$$

$$\frac{\partial y_A}{\partial \theta} = -2L_1 \sin \theta \quad \frac{\partial y_A}{\partial \varphi} = -\frac{L_2}{2} \cos \varphi; \quad \frac{\partial y_B}{\partial \varphi} = -\frac{L_2}{2} \cos \varphi$$

$$\Lambda_{ac} = \begin{bmatrix} -L_1 \cos \theta_0 & \frac{L_2}{2} \sin \phi_0 & 0 \\ -L_1 \sin \theta_0 & -\frac{L_2}{2} \cos \phi_0 & 0 \\ 1 & 0 & 0 \\ 0 & \frac{L_2}{2} & 0 \\ 0 & 1 & 0 \\ 0 & -2 \sin \phi_0 - R \cos \phi_0 & -R \cos \phi_0 \\ 0 & 2 \cos \phi_0 - R \sin \phi_0 & -R \sin \phi_0 \\ 0 & 1 & 1 \end{bmatrix} \begin{matrix} x_{q1} \\ y_{q1} \\ \theta \\ v_q \\ \phi \\ x_c \\ y_c \\ \psi_A \end{matrix}$$

$$\Lambda_c = \Lambda_{k2} = \begin{bmatrix} -2L_1 \cos \theta_0 & \frac{L_2}{2} \sin \phi_0 & 0 \\ 0 & -2 \sin \phi_0 - R \cos \phi_0 & -R \cos \phi_0 \\ 0 & \frac{3}{2} L_2 \cos \phi_0 & 0 \end{bmatrix}$$

$$\Lambda_{a2} = \begin{bmatrix} 0 & -\frac{L_2}{2} \cos \phi_0 & 0 \\ -2L_1 \cos \theta_0 & \frac{L_2}{2} \sin \phi_0 & 0 \\ -2L_1 \sin \theta_0 & -\frac{L_2}{2} \cos \phi_0 & 0 \end{bmatrix}$$

$$\frac{\Delta L_1}{\partial \theta^2} = 2L_1 \sec \theta \quad \frac{\partial^2 \Delta L_1}{\partial \phi^2} = \frac{L_2}{2} \cos \phi$$

$$\frac{\partial^2 \Delta L_2}{\partial \phi^2} = -(2-R\psi) \cos \phi + R \sec \phi$$

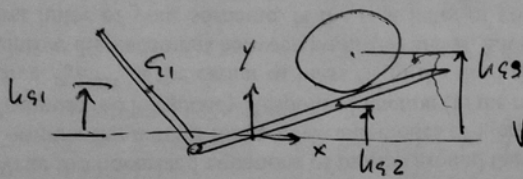
$$\frac{\partial^2 \Delta L_2}{\partial \phi \partial \psi} = R \sec \phi ; \quad \frac{\partial^2 \Delta L_2}{\partial \psi^2} = 0$$

$$\frac{\partial^2 \Delta L_3}{\partial \phi^2} = -\frac{3}{2} L_2 \sec \phi$$

$$[K_{EL, \Pi_1}] = K_1 \Delta L_{01} \begin{bmatrix} 2L_1 \sec \theta_0 & 0 & 0 \\ 0 & \frac{L_2}{2} \cos \phi_0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$[K_{EL, \Pi_2}] = K_2 \Delta L_{02} \begin{bmatrix} 0 & 0 & 0 \\ 0 & -2 \cos \phi_0 + R \sec \phi_0 & R \sec \phi_0 \\ 0 & R \sec \phi_0 & 0 \end{bmatrix}$$

$$[K_{EL, \Pi_3}] = K_3 \Delta L_{03} \begin{bmatrix} 0 & 0 & 0 \\ 0 & -\frac{3}{2} L_2 \sec \phi_0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$



$$u_{s1} = y_{s1}$$

$$u_{s2} = y_{s2} = \frac{L_2}{2} \sec \phi$$

$$u_{s3} = y_{s3}$$

$$\frac{\partial^2 u_{s1}}{\partial \theta^2} = -L_1 \cos \theta$$

$$\frac{\partial^2 u_{s1}}{\partial \phi^2} = \frac{L_2}{2} \sec \phi$$

$$\frac{\partial^2 u_{s2}}{\partial \phi^2} = -\frac{L_2}{2} \sec \phi$$

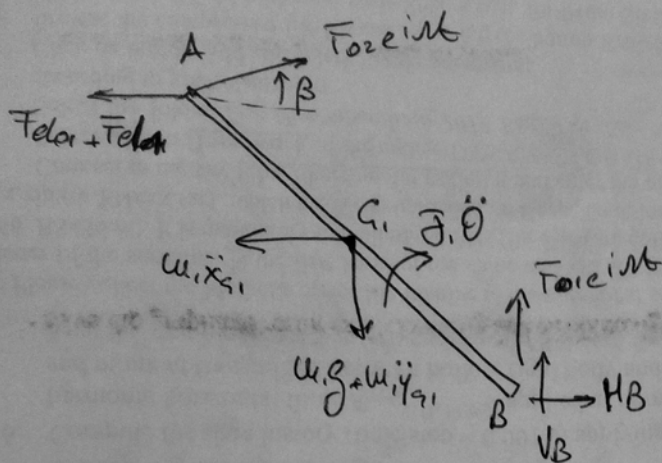
$$[K_{s1}] = u_{s1} g \begin{bmatrix} -L_1 \cos \theta_0 & 0 & 0 \\ 0 & \frac{L_2}{2} \sec \phi_0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\frac{\partial^2 u_{s3}}{\partial \phi^2} = -(2-R\psi) \sec \phi - R \cos \phi$$

$$[K_{s2}] = u_{s2} g \begin{bmatrix} 0 & 0 & 0 \\ 0 & -\frac{L_2}{2} \sec \phi_0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\frac{\partial^2 u_{s3}}{\partial \phi \partial \psi} = -R \cos \phi ; \quad \frac{\partial^2 u_{s3}}{\partial \psi^2} = 0$$

$$[K_{s3}] = u_{s3} g \begin{bmatrix} 0 & 0 & 0 \\ 0 & -2 \sec \phi_0 - R \cos \phi_0 & -R \cos \phi_0 \\ 0 & -R \cos \phi_0 & 0 \end{bmatrix}$$



$$\sum \bar{F}_x^{AB} = 0 : \quad H_B = u_1 \ddot{x}_{G_1} + \bar{F}_{el1} + F_{el1} - \bar{F}_{02} e^{int} \cos \beta$$

$$H_{B,ST} = \bar{F}_{el01}$$

$$H_{B,dyn} = u_1 \ddot{x}_{G_1} + F_{el1} - \bar{F}_{02} e^{int} \cos \beta$$

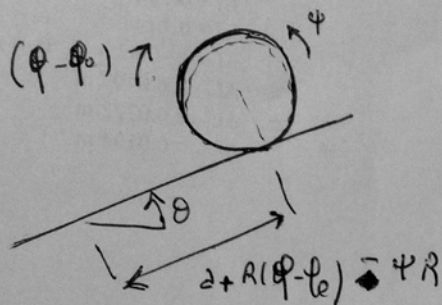
$$\sum \bar{F}_y^{AB} = 0 : \quad V_B = u_1 g + u_1 \ddot{y}_{G_1} - \bar{F}_{01} e^{int} - \bar{F}_{02} e^{int} \sin \beta$$

$$V_{B,ST} = u_1 g$$

$$V_{B,dyn} = u_1 \ddot{y}_{G_1} - \bar{F}_{01} e^{int} - \bar{F}_{02} e^{int} \sin \beta$$

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$$x = \{\theta \quad \varphi \quad \psi\}^T \quad x_0 = \{\theta_0 \quad \varphi_0 \quad \psi_0\}^T$$



ψ = rot. absolute

$$x_p = \{a + R\varphi - R\varphi_0 - \psi R\} \cos \varphi$$

$$y_p = \{a + R\varphi - R\varphi_0 - \psi R\} \sin \varphi$$

$$x_c = \{a + R\varphi - R\varphi_0 - \psi R\} \cos \varphi - R \sin \varphi$$

$$y_c = \{a + R\varphi - R\varphi_0 - \psi R\} \sin \varphi + R \cos \varphi$$

$$\frac{\partial x_c}{\partial \varphi} = -\{a + R\varphi - R\varphi_0 - \psi R\} \sin \varphi - R \cos \varphi + R \cos \varphi; \quad \frac{\partial x_c}{\partial \psi} = -R \cos \varphi$$

$$\frac{\partial y_c}{\partial \varphi} = \{a + R\varphi - R\varphi_0 - \psi R\} \cos \varphi - R \sin \varphi + R \sin \varphi; \quad \frac{\partial y_c}{\partial \psi} = -R \sin \varphi$$

$$\frac{\partial^2 \Delta z}{\partial \varphi^2} = -\{a + R\varphi - R\varphi_0 - \psi R\} \cos \varphi - R \sin \varphi; \quad \frac{\partial^2 \Delta z}{\partial \varphi \partial \psi} = R \sin \varphi$$

$$\frac{\partial^2 \Delta z}{\partial \psi^2} = -\{a + R\varphi - R\varphi_0 - \psi R\} \sin \varphi + R \cos \varphi; \quad \frac{\partial^2 \Delta z}{\partial \psi \partial \varphi} = -R \cos \varphi$$

$$\Lambda_u = \begin{bmatrix} 0 & -a \sin \varphi_0 & -R \cos \varphi_0 \\ 0 & a \cos \varphi_0 & -R \sin \varphi_0 \\ 0 & 0 & 1 \end{bmatrix} \quad \Lambda_k = \begin{bmatrix} \dots & \dots & \dots \\ 0 & -a \sin \varphi_0 & -R \cos \varphi_0 \\ \dots & \dots & \dots \end{bmatrix}$$

$$[k_{\alpha, \beta}] = k_1 \Delta \alpha \begin{bmatrix} 0 & 0 & 0 \\ 0 & -a \cos \varphi_0 - R \sin \varphi_0 & R \sin \varphi_0 \\ 0 & R \sin \varphi_0 & 0 \end{bmatrix} \quad [k_{\gamma, \delta}] = u_1 \gamma \begin{bmatrix} 0 & 0 & 0 \\ 0 & -a \sin \varphi_0 + R \cos \varphi_0 & -R \cos \varphi_0 \\ 1 & -R \cos \varphi_0 & 0 \end{bmatrix}$$